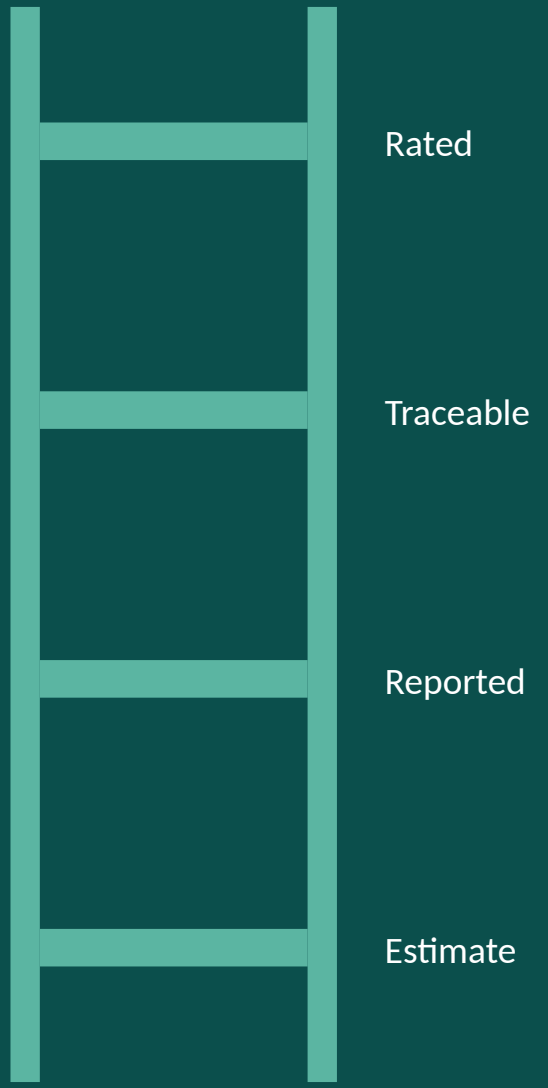
Merit Supplier Ladder

Credible data from the supply chain partners patients can't lose

Team 23 · Elijah Palmer · Cameron Boyle · Nolan Ford

June 2024 An uncleared syringe from a supplier Merit couldn't see led to a recall of Merit's custom kits.



Merit asks every supplier the same question, so the ones that matter most can't answer it

The data gap sits where risk is highest

1,030 direct suppliers in 42 countries

65% of spend is with the top 78, which are mostly already reporting

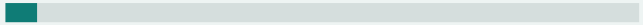
9–12 mo to requalify a supplier, so losing one means patients wait

The gap: small, single-plant makers in Mexico and China

The story is ahead of the numbers

Scope 3 unmeasured: only a 2024 baseline exists

−2% Scope 1+2 vs 2020, against a −40% goal for 2030



0 EcoVadis coverage numbers published

Merit's own 2025 report: Scope 3 from China and Mexico is a top risk

Why now

- New CEO: "responsible supply-chain practices"
- EcoVadis rollout is only a year old
- EU vague-claims ban: Sept 2026
- NHS Scope 1–3 plan: Apr 2027

Source: Case Exh. 2; Merit 2025 Sustainability Report; EU Dir. 2024/825; NHS Net Zero Supplier Roadmap

Neither default fixes the blind spot without breaking something

	A · Mandate everyone	B · Top 78 first	Ladder (ours)
Spend coverage	~ Fast on paper	✓ 65% fast	✓ 65% in month 1
Risk coverage	✗ Shallow, unverified answers	✗ Mexico/China dark for years	✓ Critical suppliers in year 1
Supplier strain	✗ High, so suppliers exit	✓ None	✓ Matched to need
HEART fit	~ Excellence only	~ Weak on Health and Responsibility	✓ All five values

89% average on unverified yes/no supplier surveys, with 40% perfect scores

56% of large Finnish firms dropped suppliers over ESG demands

Source: Case Exh. 5; research/analogues-buyer-programs.md

We sort every supplier with one rule, and each tier gets a different ask

Sort: Switching difficulty × Spend × Country risk × Readiness

	Low readiness	Moderate	High readiness
High power	Critical single-plant makers (MX/CN)	Mid-size specialty, partial reporting	Anchor partners, already reporting
Low power	Small commodity inputs	Regional distributors, packaging	Small, certified via other buyers

Merit's vendor master scores all 1,030 in weeks. The rule is the template: it works for every new supplier too. Tier counts are estimates.

Tier 1 · Anchors ~65 suppliers · ~65% of spend

We ask: Share an existing EcoVadis score (minimum 45)

We do: Accept it; no new forms. Done in month 1

Tier 2 · Critical & stuck ~30–50 · Mexico first

We ask: Readiness request → evidence pack → full EcoVadis by year 2

We do: Hands-on help: AI assistant + local consultant

Tier 3 · Long tail ~900 · easy to replace

We ask: 30-minute EcoVadis Vitals questionnaire only

We do: Merit estimates the rest from its own spend data

Source: Case Exh. 2–3; team method (backup B1)

Our AI assistant gathers the evidence, and the supplier's data never leaves the building

Stays inside the supplier's plant

Their own records

- E-invoices (CFDI, e-fapiao)
- CONTPAQi or Kingdee books
- Utility bills, logs

Local consultant

Knows the supplier's software. Runs Merit's AI assistant on-site, on the supplier's machines.

Evidence pack

kWh, fuel, policies, gaps, and a source for every figure.

Supplier approves, then it's shared

- To EcoVadis for the rating
- To Merit for Scope 3
- Raw data stays with the supplier

Each pack moves the supplier up a rung

0 · Estimate spend-based → **1 · Reported** Vitals + first numbers → **2 · Traceable** Evidence pack approved → **3 · Rated** Full EcoVadis score

The assistant only finds records that exist. If something was never measured, the gap becomes a plan to start measuring it.

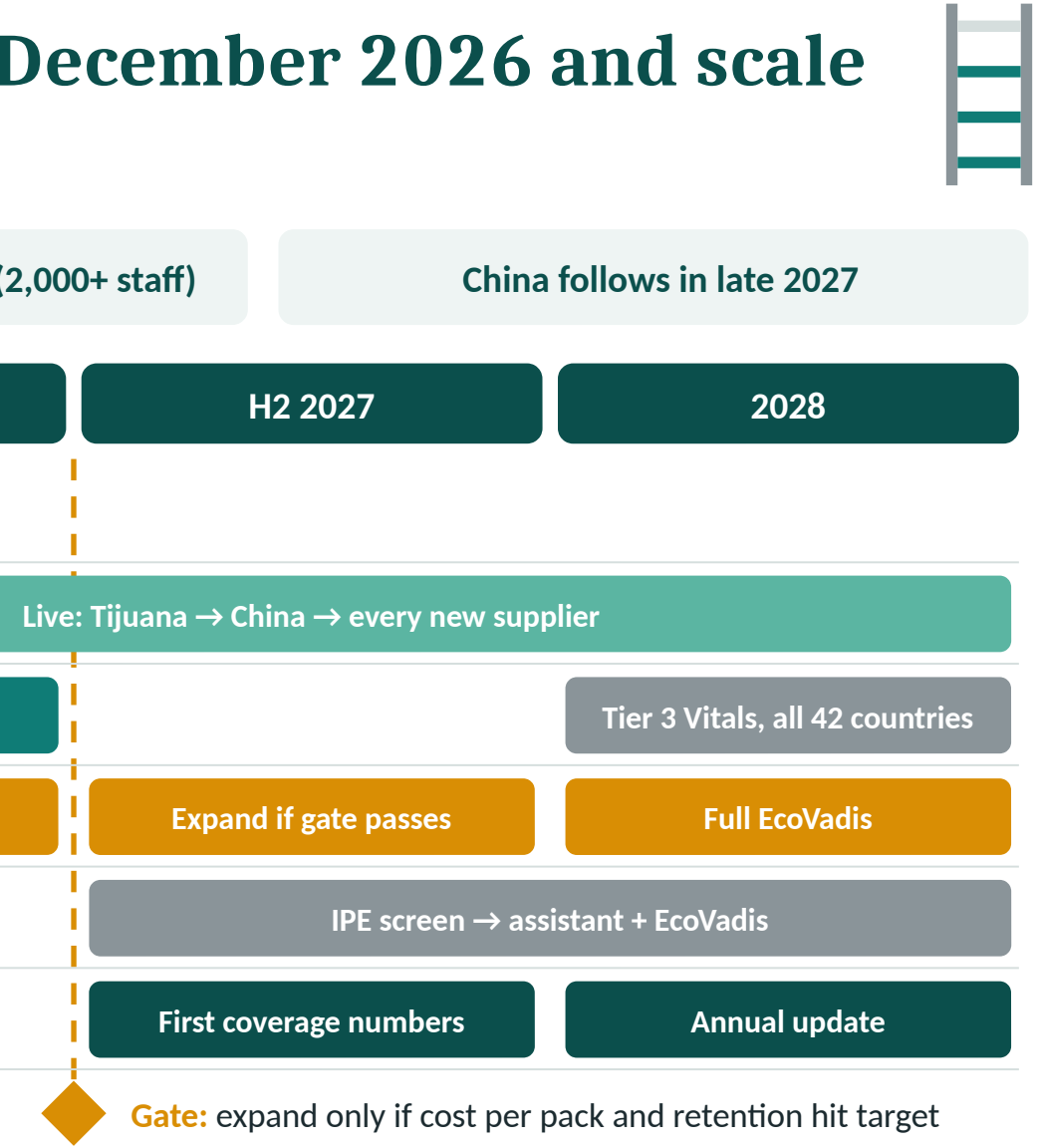
Source: SAT CFDI e-invoicing; EcoVadis evidence rules; design detail in backup B9

We pilot with 12 Tijuana suppliers in December 2026 and scale only when results hold

12 sole-source suppliers, picked by the rule

Beside Merit's largest plant (2,000+ staff)

China follows in late 2027



Owners VP ESG (sponsor) · Procurement · Sustainability data · IT (assistant) · Finance · Comms

Source: Merit 2025 Sustainability Report p.23; team plan (pilot detail in backup B4)

Is it feasible? About $0.8M in year one, and it pays back by catching one failure every four years

$0.8M

in year one, capped, including a ~$150k AI build: ~0.26% of non-GAAP operating income

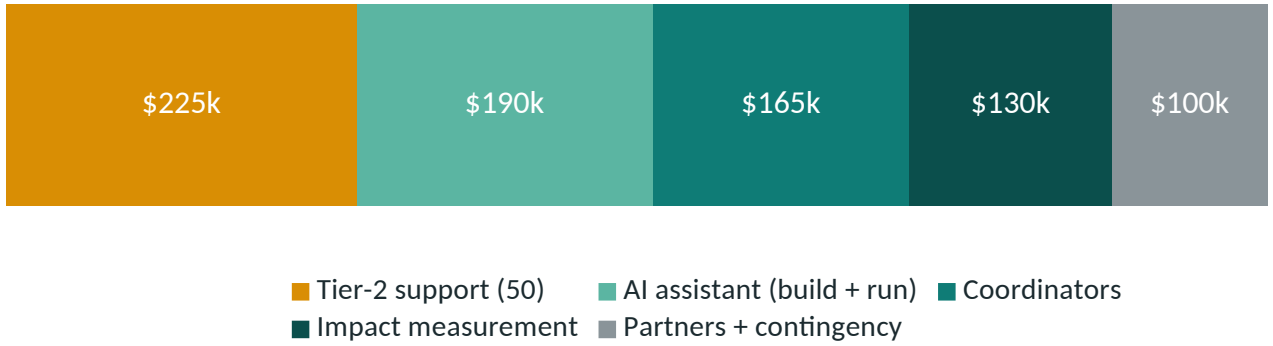
1 in 4 yrs

Breakeven at ~$0.66M a year after: catch one ~$2.5M supplier failure earlier (est.)

38%

of a year's earnings lost to supply shocks per decade (medtech average)

Year-one cost build (estimate, $k)



Margin target holds
20–22% target; Q2 2026 was 22.6%

Cash target already met
Free cash flow ≈ $401M vs $400M goal

Built once, reused
Each new supplier costs consultant hours, not a new build

Source: Merit 2025 10-K, Q2 2026 results; McKinsey medtech resilience; team cost model (backup B3)

Suppliers get paid faster, audited less, and win other buyers



Merit pays the assessment

Tier 2 EcoVadis invitation and subscription covered

Paid faster

Early payment through NAFIN's fee-free reverse factoring, for suppliers who finish

Longer contracts

Multi-year terms for completers, through normal quality and procurement review

Audited less

One EcoVadis score serves all their customers, instead of 3–5 audits a year

Help with the 40-hour week

Mexico's 48→40-hour reform is their 2026 problem; we help them comply without cutting pay

Win other buyers

U.S. purchasing groups, the NHS and French hospitals ask for this data too

Source: CDP/HSBC 2024; WRAP 2023 audit survey; NAFIN Cadenas Productivas; research/customer-esg-demands.md

Success means covering the suppliers patients depend on, and claiming only what we can prove



Metric	2027	2028	Benchmark
Direct spend with a rating	**85%**	**90%**	Stryker: 85% target by 2027
Critical suppliers with a traceable evidence pack	**60%**	**90%**	**No peer reports this**
Tier 2 supplier retention	**100%**	**100%**	Attrition tripwire
Pilot workers inside the legal week, with weekly pay held	**Baseline**	**Rising**	vs a wait-listed comparison group
Supplier-caused backorders on Tier 2 parts	**→ 0**	**0**	FDA 506J shortage notices

What Merit publishes each year: coverage by spend and by critical supplier, and how much of Scope 3 is real supplier data vs estimate. No "climate neutral" or vague claims (EU ban, Sept 2026; FTC Green Guides).

Source: Stryker 2024 proxy; research/impact-measurement.md; communication plan in backup B10

Others have done this, and it worked

700+

Schneider Electric

suppliers measured their carbon footprint for the first time; 950 now disclose publicly

120+

Philips × Zhejiang Haishun (China)

improvements found by partnering with the supplier instead of auditing it

52%

CDP & HSBC, 23,000 suppliers

more likely to cut emissions when buyers offer financial incentives, not just training

22%

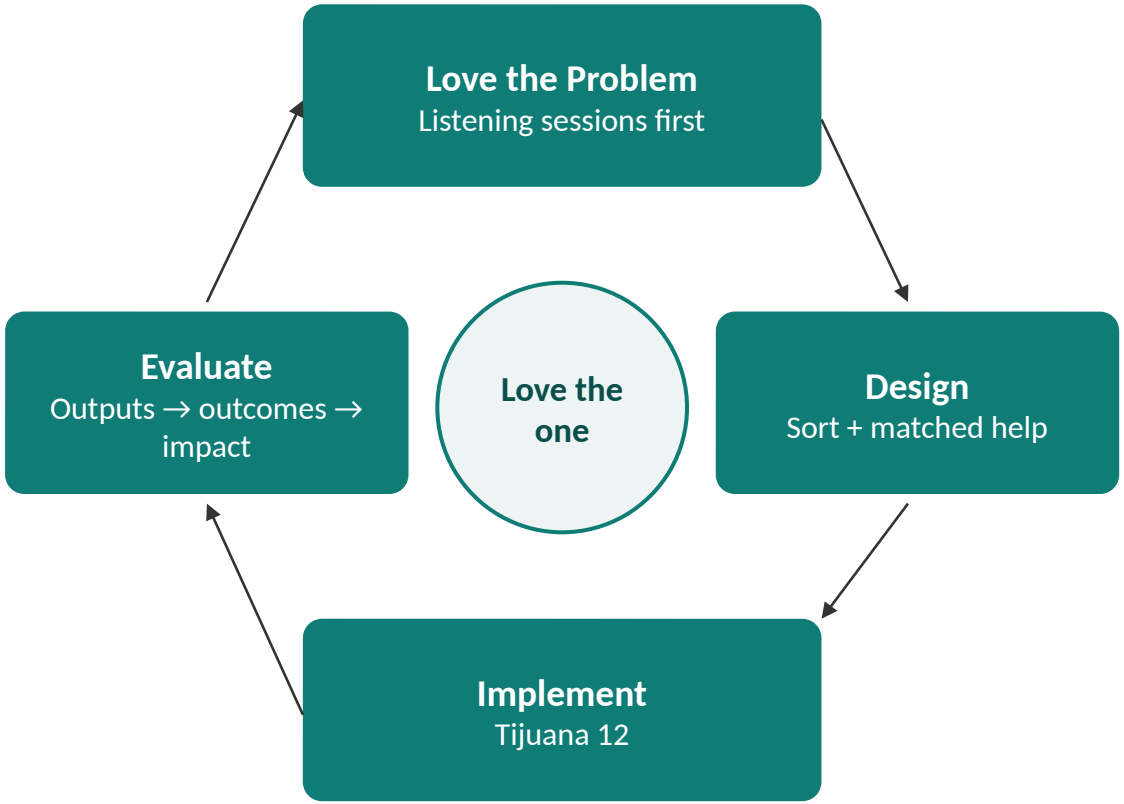
Better Work (ILO / IFC)

more productive factory lines after supervisors were trained

Peers set the bar: Stryker targets 85% of direct spend assessed by 2027; Medtronic and Philips tier their support by supplier maturity, as we do.

Source: Schneider Electric 2025; Philips Dec 2021; CDP/HSBC 2024; Better Work (Tufts); full table in backup B2

Built on the Social Impact Cycle, with tripwires for what could go wrong



Harm → tripwire

- **Dependency on Merit's help** → time-boxed, capped support
- **Paperwork without practice** → third-party evidence review
- **Supplier exits or passes on cost** → retention 100%; no Y1 penalties
- **Hour cuts lower take-home pay** → weekly pay must hold
- **Supplier data exposed** → stays on-site; supplier approves

Unmitigated harm → Mitigating harm → Internal initiatives → Offsets / philanthropy → Prosocial marketing → **Socially strategic**

Source: Ballard Center Social Impact Cycle and Journey; Wharton on TOMS

Carrying Merit's mission, vision and values down the supply chain

Vision "The most customer-focused company in healthcare." That customer is a patient waiting on a device.

Understand
Ask suppliers what's blocking them first

Innovate
An AI assistant that keeps their data in their hands

Deliver
Critical parts keep reaching patients

Mission

H — **Health**
Workers and patients protected

E — **Excellence**
Same bar for every partner

A — **Agility**
Country risk in the rule

R — **Responsibility**
A real Scope 3 baseline

T — **Teamwork**
We listen first

HEART

The ask

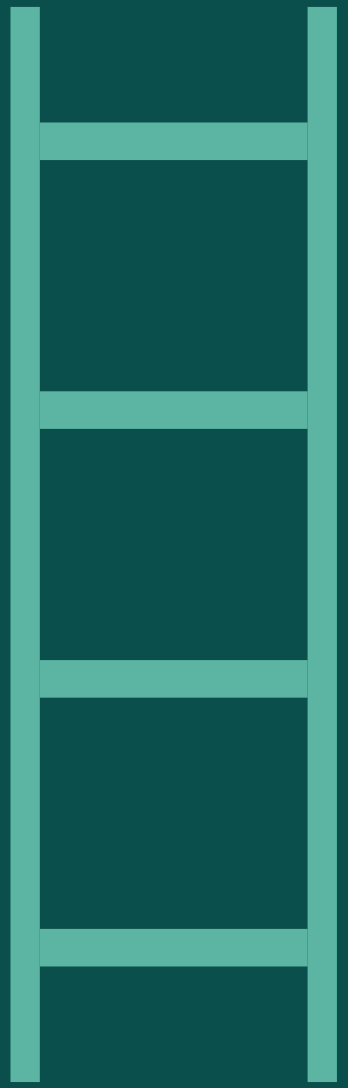
1. **Approve the sorting rule**
2. **Fund ~$0.8M for year one, capped**
3. **Start Tijuana in Dec 2026**

The next Shenli gets caught earlier.

Source: Merit Vision & Values

Questions

Sort every supplier with one rule, then help each one climb



B1 · How the sorting rule works

Factor	Scores 1 (low)	Scores 3 (high)
Switching difficulty	Many qualified alternates	Sole-source; 9–12 months to requalify
Spend	Bottom of the spend list	In the top 78
Country risk	Low-risk country	Mexico, China (Merit's own climate-risk research)
Readiness	Already rated on EcoVadis	No ESG staff, no emissions tracking

Priority = Switching × Spend × Country risk (1–27). Tier 2 = priority ≥ 8 and readiness low/moderate · Tier 1 = ready (already rated) · Tier 3 = everyone else. Thresholds are illustrative.

Example (illustrative)	Switch	Spend	Country	Priority	Readiness	Tier
Tijuana molded-part maker, sole-source	3	2	3	**18**	Low	Tier 2
Large public US resin supplier	2	3	1	**6**	High	Tier 1
Regional packaging supplier	1	1	2	**2**	Moderate	Tier 3

Source: Case Exh. 2–3; team method

B2 · What peers do

Company	Approach	Result or target
Stryker	Third-party ESG assessments	85% of direct spend by 2027 (~59% in 2024)
BD	Desktop ESG assessments	76% of spend assessed (FY24); goal 90% of eligible spend
Abbott	Risk screening + due diligence	>90% of direct-material spend screened
J&J	EcoVadis for program suppliers + CDP disclosure	Assessment and emissions kept as separate asks
Philips	Four maturity categories, tailored support	504 suppliers in CDP program, 88% response
Medtronic	Playbook: calculate → target → report → reduce	Support modules by supplier maturity
Schneider	Zero Carbon Project, ~1,000 suppliers	700+ measured a footprint for the first time; 950 disclose

Source: Company sustainability reports and filings; research/outside-research.md; Cameron's benchmark file

B3 · Cost build and sensitivity

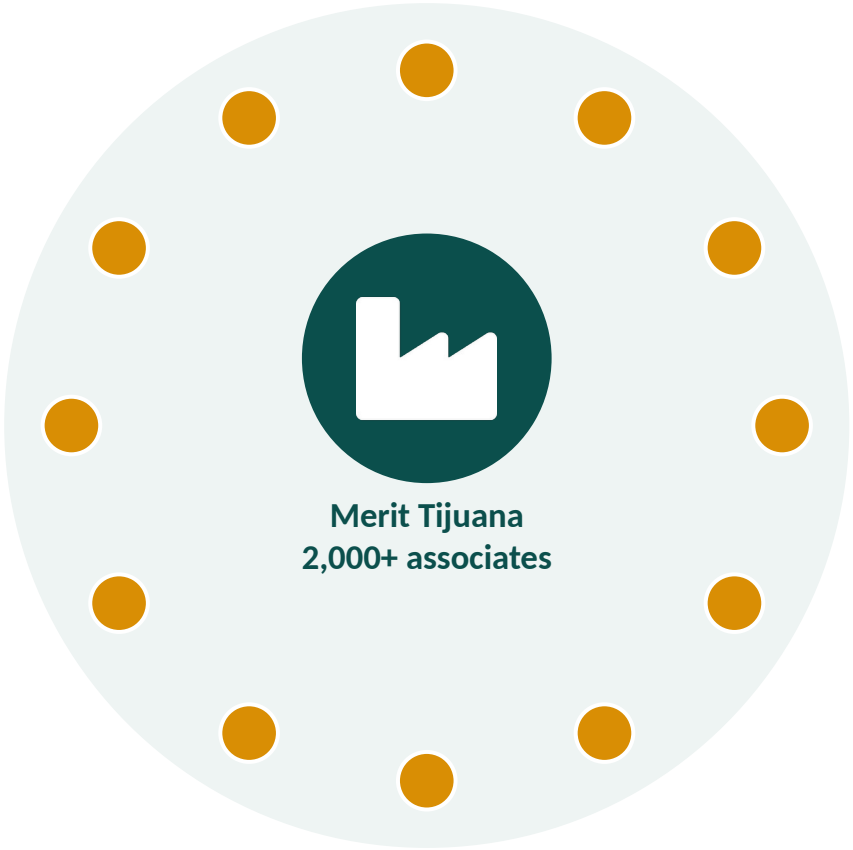
Line item	Basis	Year 1 $k
Tier-2 support	50 × ~$4.5k (subscription + consultant hours)	225
AI assistant	~$150k one-time build + ~$40k/yr to run	190
Coordinators	1 Utah + 1 Tijuana, loaded	165
Impact measurement	Worker survey, IMSS data, comparison group	130
Partners + contingency	Translation, workshops, buffer	100
Total	Year 1 (run-rate after: ~660)	**~810**

Tier 2 size	Run-rate / yr	% op. income
25 suppliers	$0.55M	0.18%
50 suppliers	**$0.66M**	**0.21%**
100 suppliers	$0.89M	0.29%

- Cross-check: Nolan's bottom-up model ≈ $73k for 12 intensive suppliers (~$6k each) vs our ~$4.5k
- Breakeven: ~$0.66M ÷ ~$2.5M per sole-source failure (assumption) ≈ one caught earlier every 4 years
- Scale of a failure: Merit revenue ≈ $4.15M a day; Baxter's one hurricane-hit plant ≈ $200M of lost sales
- AI build is an estimate (off-the-shelf AI platform + integration); get a vendor quote. Op income ~$308M (non-GAAP, 2025)

Source: research/financials-value.md; supplier-support-budget-and-decisions.md

B4 · Tijuana pilot detail



● 12 pilot suppliers, chosen by the rule (positions illustrative)

Who
12 sole-source suppliers near Merit's largest plant; ~73 device companies in the cluster

Delivery
A coordinator in Procurement; local consultants who know CONTPAQi run the AI assistant

Model
IFC's Mexico supplier development program (automotive today; we ask for a medtech track)

Bonuses, not load-bearing
Baja California certification reimbursement; medical-device cluster workshops

Source: Merit 2025 Sustainability Report p.23; IFC–INA–SE release Aug 2025; research/analogues-mexico-china.md

B5 · China plan (phase 2, 2027)

Screen now: IPE Blue Map
Beijing NGO database of government and company environmental records. Flags public violations; it is not a carbon inventory or a labor audit.

Then: AI assistant + local consultant
Same model as Mexico: a consultant who knows Kingdee or Yonyou runs it on e-fapiao and books. EcoVadis already works in Chinese.

Model: Philips × Zhejiang Haishun
Collaboration over auditing; 120+ improvement opportunities found (savings were projected).

Not RBA
Placed on China's MOFCOM countermeasure list (Aug 2026). Single source; verify before citing.

UFLPA screening from day 1
Check every China supplier against the entity list.

Source: IPE; Philips Dec 2021; EcoVadis language support; research/analogues-mexico-china.md

B6 · EcoVadis mechanics

Vitals

- ≤30-minute questionnaire, under 20 questions, 14 languages
- Risk profile; the on-ramp for small suppliers
- Price needs a quote

Full rating

- Environment, labor and human rights, ethics, sustainable procurement
- Evidence reviewed by EcoVadis analysts; documents made only for the questionnaire are rejected
- Score 0–100; we ask for ≥45

Carbon Action Manager

- Supplier emissions figures with reliability levels
- Feeds Scope 3; a rating is not emissions data
- Confirm Merit's subscription and exports

Source: EcoVadis support center; supplier-emissions-data-and-ecovadis.md

B7 · Mexico specifics

Topic	What's true	How we use it
IFC–INA–SE supplier program	Automotive anchor-firm program (Aug 2025): diagnostics, training, mentoring	Model it; ask IFC for a medtech track. Access unconfirmed
NAFIN Cadenas Productivas	Fee-free reverse factoring priced off the buyer's credit. SME loans ~14.4% vs TIIE 6.76%	Early payment for completers. The ESG link is our design
Labor reform	Legal week 48 → 46 hrs (2027) → 40 hrs (2030)	Help suppliers comply; tripwire: weekly pay holds
E-invoicing	Every invoice is a CFDI XML file filed with SAT, including CFE electricity bills	The AI assistant's main data source
Local reporting	SEMARNAT RENE (threshold-based), NIF/NIS, CNBV: applies case by case	Check which applies per supplier; none replaces EcoVadis

Source: research/analogues-mexico-china.md; supplier-emissions-data-and-ecovadis.md; SAT

B8 · Scope 3 method: estimate first, then replace with supplier data

Spend-based (now)
Spend × sector emission factor. Labeled as an estimate.

Hybrid (2027)
Supplier activity data (kWh, fuel) from evidence packs for critical suppliers.

Supplier-specific
Traceable packs, allocated to what Merit actually buys.

- A supplier's whole footprint isn't Merit's: allocate by what Merit buys
- Missing data isn't zero: keep the estimate until a supplier figure replaces it
- Report the mix every year: % from supplier data vs % estimated
- A traceable calculation can be checked; independent assurance is a separate, later step

Source: GHG Protocol Scope 3 Category 1 guidance; supplier-emissions-data-and-ecovadis.md

B9 · The AI evidence assistant: how it works and why suppliers can trust it

What it reads

- Mexico: CFDI e-invoices (electricity, fuel, gas), CONTPAQi / Aspel exports
- China: e-fapiao, Kingdee / Yonyou exports
- Production logs, payroll hour summaries, permits and certificates

What it produces

- Activity data: kWh, liters of fuel, refrigerants
- Policies and practices found, with the document behind each
- A gap list and a plan to start collecting what's missing

Safeguards

- Runs on the supplier's machines; raw data never leaves
- Local consultant works for the supplier, under NDA
- Every figure cites its source; consultant checks it; supplier approves the pack
- Can't invent records: EcoVadis rejects documents made only for the questionnaire

Ownership: Merit owns the tool; the supplier owns its data. Build ~$150k once on an off-the-shelf AI platform (estimate; get a quote), ~$40k/yr to run.

Source: EcoVadis evidence rules; SAT CFDI; team design

B10 · What Merit says, to whom, and when

Audience	What Merit shares	When
Suppliers	Their tier, the ask, the support, the timeline, in their language	Dec 2026
Hospitals & GPOs	Scope 1/2 + plan; coverage by spend and by critical suppliers	2027 · NHS Scope 1–3 plan by Apr 2027
Board & investors	Cost vs cap, coverage KPIs, risk	Quarterly
Associates	Procurement and Quality roles in the program	Launch + annual report

Guardrails

- Publish what isn't known yet (Scope 3: % supplier data vs % estimate)
- No "climate neutral" or vague claims (EU ban, Sept 2026; FTC Green Guides)
- "Working with our partners," never "our suppliers are sustainable"

Source: EU Dir. 2024/825; FTC Green Guides; NHS Net Zero Supplier Roadmap

B11 · Sources

- Merit Medical case, BFG 2026 (Exhibits 2, 3, 5)
- Merit 2025 Sustainability Report; 2025 10-K; Q2 2026 results
- Merit recall notice (Jun 11 2024); FDA recall record 94797
- EU Directive 2024/825 (empowering consumers); FTC Green Guides
- NHS Net Zero Supplier Roadmap
- CDP/HSBC, Strengthening the Chain (2024)
- McKinsey, medtech supply-chain resilience
- Stryker 2024 proxy; BD, Abbott, J&J, Philips, Medtronic sustainability reports
- Schneider Electric Zero Carbon Project
- GHG Protocol Scope 3 guidance
- EcoVadis Vitals guide, rating methodology, evidence rules
- IFC–INA–SE supplier program release (Aug 2025)
- NAFIN Cadenas Productivas; SAT CFDI e-invoicing
- WRAP audit-burden survey (2023); Better Work (Tufts)
- Ballard Center Social Impact Cycle; Wharton on TOMS
- Full list: research/sources.txt in the team repo